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An introduction to Topological Data Analysis: fundamental and practical aspects for data scientists

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Abstract

Topological Data Analysis (TDA) is a recent and fast growing field providing a set of new topological and geometric tools to infer relevant features for possibly complex data. This paper is a brief introduction, through a few selected topics, to basic fundamental and practical aspects of TDA for non experts.

1 Introduction and motivation

Topological Data Analysis (TDA) is a recent field that emerged from various works in applied (algebraic) topology and computational geometry during the first decade of the century. Although one can trace back geometric approaches for data analysis quite far in the past, TDA really started as a field with the pioneering works of Edelsbrunner et al. (2002) and Zomorodian and Carlsson (2005) in persistent homology and was popularized in a landmark paper in 2009 Carlsson (2009). TDA is mainly motivated by the idea that topology and geometry provide a powerful approach to infer robust qualitative, and sometimes quantitative, information about the structure of data - see, e.g. Chazal (2017).

TDA aims at providing well-founded mathematical, statistical and algorithmic methods to infer, analyze and exploit the complex topological and geometric structures underlying data that are often represented as point clouds in Euclidean or more general metric spaces. During the last few years, a considerable effort has been made to provide robust and efficient data structures and algorithms for TDA that are now implemented and available and easy to use through standard libraries such as the Gudhi library [1] (C++ and Python) Maria et al. (2014) and its R software interface Fasy et al. (2014a), Dionysus[2], PHAT[3], DIPHA[4], or Giotto [5]. Although it is still rapidly evolving, TDA now provides a set of mature and efficient tools that can be used in combination or complementary to other data sciences tools.

The TDA pipeline. TDA has recently known developments in various directions and application fields. There now exist a large variety of methods inspired by topological and geometric approaches. Providing a complete overview of all these existing approaches is beyond the scope of this introductory survey. However, many standard ones rely on the following basic pipeline that will serve as the backbone of this paper:

1. The input is assumed to be a finite set of points coming with a notion of distance - or similarity - between them. This distance can be induced by the metric in the ambient space (e.g. the Euclidean metric when the data are embedded in $\mathbb{R}^d$) or come as an intrinsic

[1] https://gudhi.inria.fr/
[2] http://www.mrzv.org/software/dionysus/
[3] https://bitbucket.org/phat-code/phat
[4] https://github.com/DIPHA/dipha
[5] https://giotto-ai.github.io/gtda-docs/0.4.0/library.html

metric defined by a pairwise distance matrix. The definition of the metric on the data is usually given as an input or guided by the application. It is however important to notice that the choice of the metric may be critical to reveal interesting topological and geometric features of the data.

2. A "continuous" shape is built on top of the data in order to highlight the underlying topology or geometry. This is often a simplicial complex or a nested family of simplicial complexes, called a filtration, that reflects the structure of the data at different scales. Simplicial complexes can be seen as higher dimensional generalizations of neighboring graphs that are classically built on top of data in many standard data analysis or learning algorithms. The challenge here is to define such structures that are proven to reflect relevant information about the structure of data and that can be effectively constructed and manipulated in practice.

3. Topological or geometric information is extracted from the structures built on top of the data. This may either results in a full reconstruction, typically a triangulation, of the shape underlying the data from which topological/geometric features can be easily extracted or, in crude summaries or approximations from which the extraction of relevant information requires specific methods, such as e.g. persistent homology. Beyond the identification of interesting topological/geometric information and its visualization and interpretation, the challenge at this step is to show its relevance, in particular its stability with respect to perturbations or presence of noise in the input data. For that purpose, understanding the statistical behavior of the inferred features is also an important question.

4. The extracted topological and geometric information provides new families of features and descriptors of the data. They can be used to better understand the data - in particular through visualization- or they can be combined with other kinds of features for further analysis and machine learning tasks. These information can also be used to design well-suited data analysis and machine learning models. Showing the added-value and the complementarity (with respect to other features) of the information provided by TDA tools is an important question at this step.

TDA and statistics. Until quite recently, the theoretical aspects of TDA and topological inference mostly relied on deterministic approaches. These deterministic approaches do not take into account the random nature of data and the intrinsic variability of the topological quantity they infer. Consequently, most of the corresponding methods remain exploratory, without being able to efficiently distinguish between information and what is sometimes called the "topological noise".

A statistical approach to TDA means that we consider data as generated from an unknown distribution, but also that the inferred topological features by TDA methods are seen as estimators of topological quantities describing an underlying object. Under this approach, the unknown object usually corresponds to the support of the data distribution (or part of it). The main goals of a statistical approach to topological data analysis can be summarized as the following list of problems:

Topic 1: proving consistency and studying the convergence rates of TDA methods.

Topic 2: providing confidence regions for topological features and discussing the significance of the estimated topological quantities.

Topic 3: selecting relevant scales at which the topological phenomenon should be considered, as a function of observed data.

Topic 4: dealing with outliers and providing robust methods for TDA.

Applications of TDA in data science. On the application side, many recent promising and successful results have demonstrated the interest of topological and geometric approaches in an increasing number of fields such has, e.g., material science Kramar et al. (2013); Nakamura et al. (2015) 3D shape analysis Skraba et al. (2010); Turner et al. (2014b), image analysis Qaiser et al. (2019); Rieck et al. (2020), multivariate time series analysis Khasawneh and Munch (2016); Seversky et al. (2016); Umeda (2017), medicine Dindin et al. (2020), biology Yao et al. (2009) , genomic Carrière and Rabadán (2020) chemistry Lee et al. (2017) or sensor networks De Silva and Ghrist (2007) to name a few. It is beyond the scope to give an exhaustive list of applications of TDA. On another hand, most of the successes of TDA result from its combination with other analysis or learning techniques - see Section 6.5 for a discussion and references. So, clarifying the position and complementarity of TDA with respect to other approaches and tools in data science is also an important question and an active research domain.

The overall objective of this survey paper is two-fold. First, it intends to provide data scientists with a brief and comprehensive introduction to the mathematical and statistical foundations of TDA. For that purpose, the focus is put on a few selected, but fundamental, tools and topics: simplicial complexes (Section 2) and their use for exploratory topological data analysis (Section 3), geometric inference (Section 4) and persistent homology theory (Section 5) that play a central role in TDA. Second, this paper also aims at demonstrating how, thanks to the recent progress of software, TDA tools can be easily applied in data science. In particular, we show how the Python version of the Gudhi library allows to easily implement and use the TDA tools presented in this paper (Section 7). Our goal is to quickly provide the data scientist with a few basic keys - and relevant references - to get a clear understanding of the basics of TDA to be able to start to use TDA methods and software for his own problems and data.

2 Metric spaces, covers and simplicial complexes

As topological and geometric features are usually associated to continuous spaces, data represented as finite sets of observations, do not directly reveal any topological information per se. A natural way to highlight some topological structure out of data is to "connect" data points that are close to each other in order to exhibit a global continuous shape underlying the data. Quantifying the notion of closeness between data points is usually done using a distance (or a dissimilarity measure), and it often turns out to be convenient in TDA to consider data sets as discrete metric spaces or as samples of metric spaces.

Metric spaces. Recall that a metric space (M, ρ) is a set M with a function $\rho : M \times M \to \mathbb{R}_+$, called a distance, such that for any $x, y, z \in M$:
i) $\rho(x, y) \geq 0$ and $\rho(x, y) = 0$ if and only if $x = y$,
ii) $\rho(x, y) = \rho(y, x)$ and,
iii) $\rho(x, z) \leq \rho(x, y) + \rho(y, z)$.
Given a a metric space (M, ρ), the set $\mathcal{K}(M)$ of its compact subsets can be endowed with the so-called *Hausdorff distance*: given two compact subsets $A, B \subseteq M$ the Hausdorff distance $d_H(A, B)$ between A and B is defined as the smallest non negative number δ such that for any $a \in A$ there exists $b \in B$ such that $\rho(a, b) \leq \delta$ and for any $b \in B$, there exists $a \in A$ such that $\rho(a, b) \leq \delta$ - see Figure 1. In other words, if for any compact subset $C \subseteq M$, we denote by $d(., C) : M \to \mathbb{R}_+$ the distance function to C defined by $d(x, C) := \inf_{c \in C} \rho(x, c)$ for any $x \in M$, then one can prove that the Hausdorff distance between A and B is defined by any of the two following equalities:

$$\begin{aligned} d_H(A, B) &= \max\{\sup_{b \in B} d(b, A), \sup_{a \in A} d(a, B)\} \\ &= \sup_{x \in M} |d(x, A) - d(x, B)| = \|d(., A) - d(., B)\|_\infty \end{aligned}$$